25901096 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBIF201 - GENOMICS & COMPUTATIONAL GENOME ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Analyze the role of mutations in evolution. [An] / [CO4]
2. Adore the role of ethics in protecting individuals during genome sequencing. [Ap] / [CO3]
3. Identify the role of radiation in radiation hybrid mapping. [U] / [CO2]
4. Explain what is meant by gene expression. [U] / [CO1]
5. Use BLAST in gene prediction. [A] / [CO7]
6. Celebrate the role of genome annotation. [Ap] / [CO6]
7. Execute any two steps in variant analysis. [A] / [CO7]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

8. Describe mitochondrial genome and list its main features. [K] / [CO4]

9. Celebrate the historical importance of launching the Human Genome Project in 1990. [Ap] / [CO3]

10. Commend the steps in microarray method. [Ap] / [CO6]

11. Execute comparison between ORF Finder and GenScan. [A] / [CO7]

12. Salute the role of bioinformatics tools in RNA-Seq. [Ap] / [CO6]

13. Identify with the contribution of whole genome shotgun sequencing to modern genomics. [Ap] / [CO3]

[5x4 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

14. Describe how centi-Morgan and LOD score are related to gene mapping. [U] / [CO2]

15. Value genome assembly as a key step in understanding gene organization and function. [Ap] / [CO3]

16. Compile advantages and limitations of different assembly methods. [A] / [CO7]

17. Discover functional RNA's and its importance in detail. [A] / [CO5]

[10x2 = 20]